Cpts 317. 11/2 (makeup on 11/3)

Today: Grammar to Machine.

Thm. c.f. languages are exactly the languages accepted by PDA.

Proof. two parts:

① Grammar → machine
(c.f. grammar) (PDA)

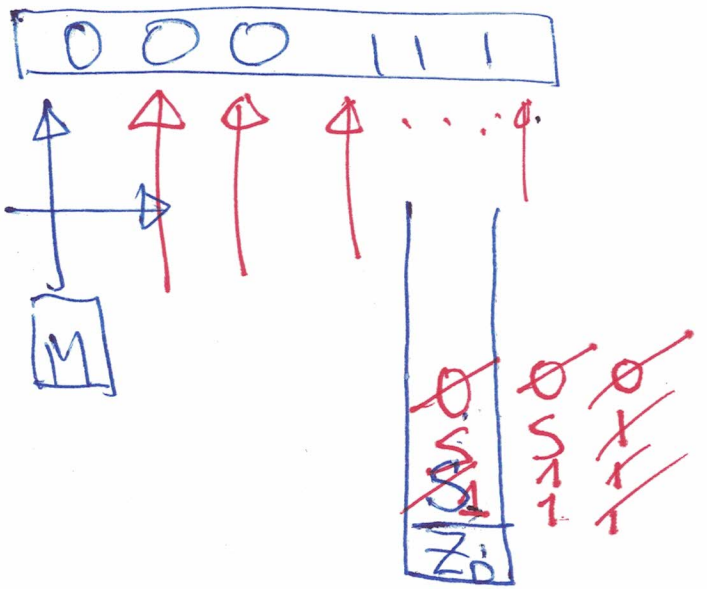
② machine → Grammar.
(PDA) (c.f. grammar).

Example. Grammar $S \to 0S1 \mid 01$.

$S \Rightarrow 0S1$

$\Rightarrow 00S11$

$\Rightarrow 000111$



Key: run the grammar rules inside the stack.

I apply $S \longrightarrow 0S1$

by (inside stack) replacing top of stack symbol S with $0S1$.

↑ new top of stack.

Summary:

$$\delta(q_0, \Lambda, Z_0) = \{(q, SZ_0)\}$$

// Initially, we push S onto the empty stack

$$\delta(q, \Lambda, Z_0) = \{(q_1, Z_0)\}$$

// q_1 is accepting state.

$$\delta(q, a, a) = \{(q, \Lambda)\}, \text{ for each } a \in \Sigma.$$

input symbol

top of stack

each terminal symbol in G.

$$\delta(q, \Lambda, A) = \{(q, \alpha) : \underbrace{A \to \alpha}_{\text{a grammar rule.}} \text{ in } G\}.$$

read nothing from input

top of stack, also a nonterminal symbol in G

Given: grammar G

$$S \longrightarrow 0S1 \mid 01$$

I construct a PDA:

$$\delta(q, *, *) = \{(q, \Lambda)\}, \quad * \in \{0,1\};$$

$$\delta(q_0, \Lambda, Z_0) = \{(q, SZ_0)\}$$

$$\delta(q, \Lambda, Z_0) = \{(q_1, Z_0)\}$$

accepting

$$\delta(q, \Lambda, S) = \{(q, 0S1), (q, 01)\}$$

// corresponding to $S \to 0S1 \mid 01$

Given: the following grammar G

$$S \rightarrow aB \mid bA$$

$$A \rightarrow a \mid aS \mid bAA$$

$$B \rightarrow b \mid bS \mid aBB$$

Want: convert G into PDA.

$$\delta(q_0, \Lambda, Z_0) = \{(q, SZ_0)\}$$

initial

$$\delta(q, *, *) = \{(q, \Lambda)\}, \quad * \in \{a, b\}$$

$$\delta(q, \Lambda, S) = \{(q, aB), (q, bA)\}$$

$$\delta(q, \Lambda, A) = \{(q, a), (q, aS), (q, bAA)\}$$

$$\delta(q, \Lambda, B) = \{(q, b), (q, bS), (q, aBB)\}$$

$$\delta(q, \Lambda, Z_0) = \{(q_1, Z_0)\}$$

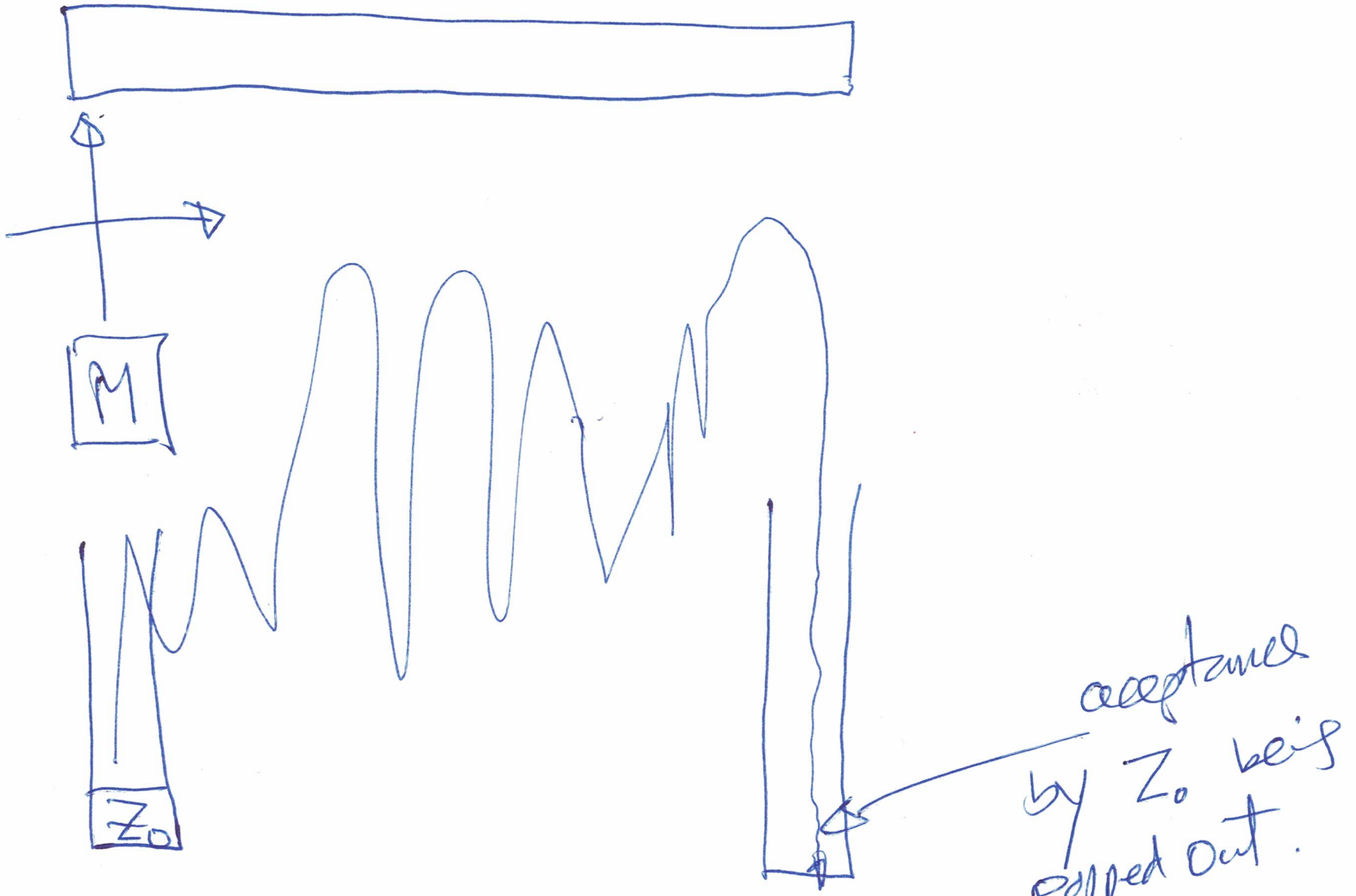
accepting

Summary: We've done with grammar $\longrightarrow$ machine.

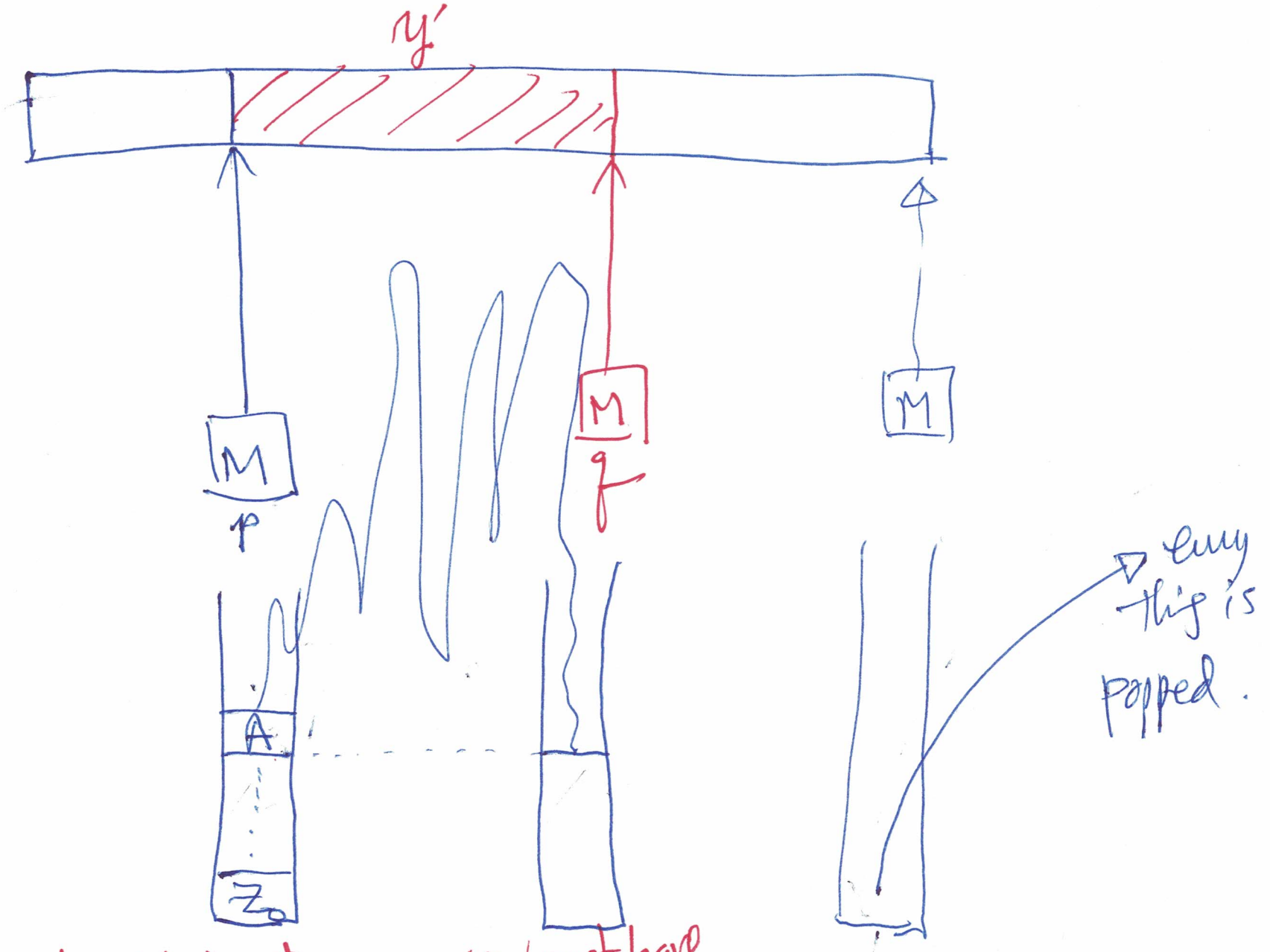
Machine $\longrightarrow$ Grammar.

Purpose: Given a PDA M, I want a c.f. grammar G s.t. $\underbrace{L(M)}_{\text{lang accepted by } M} = \underbrace{L(G)}_{\text{lang generated by grammar } G}$

understand the intuition first.



WLOG, assume that when M accepts at the end of the input, the stack is absolutely empty (even Z_0 is out)



at this moment, M is at state p, top of stack is A.

you must have a moment when the A is popped out.

y' is the "price" we pay to eventually pop the A out while moving from p to q:

$$[p, A, q] \Rightarrow_G^* y'$$

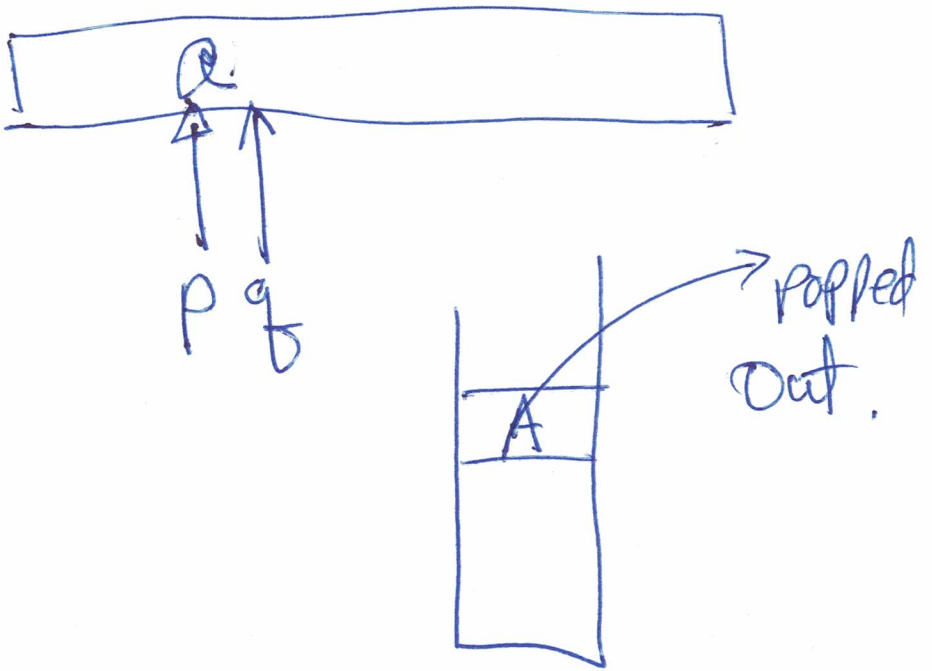
"one" nonterminal symbol in grammar.

(our intuition).

We will translate each instruction in M into grammar rule(s):

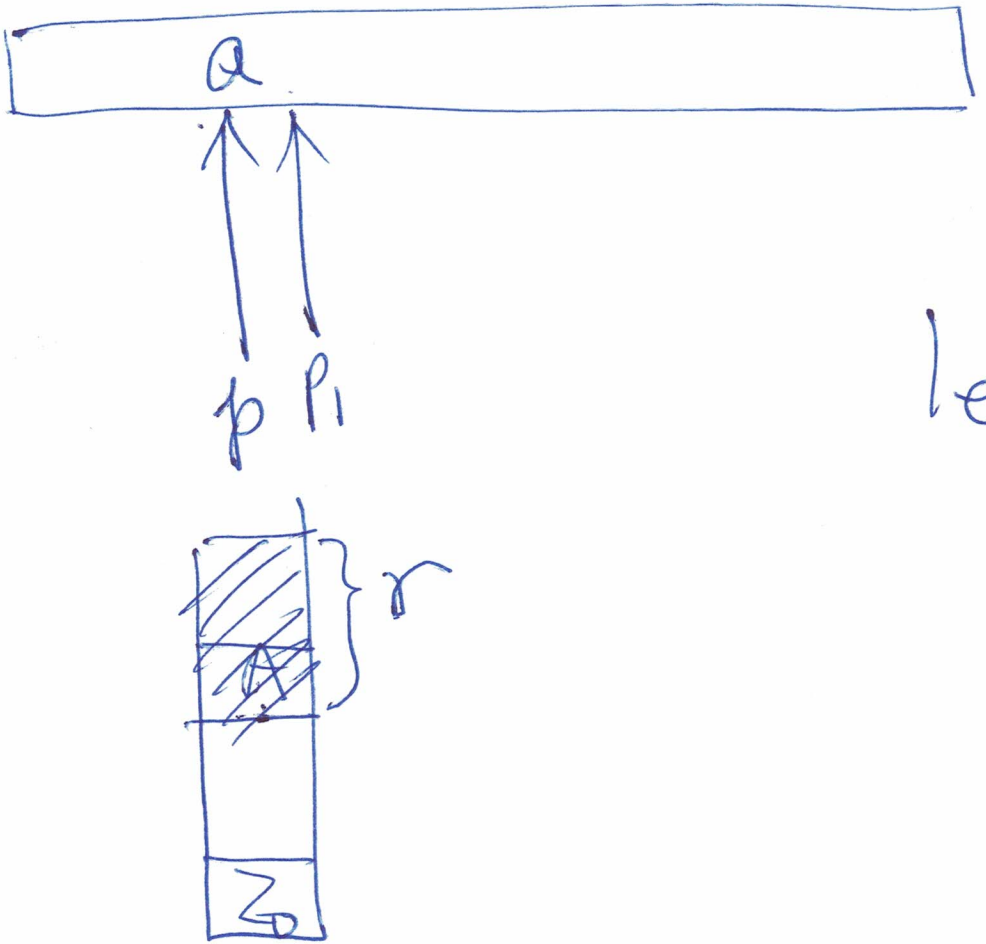
Alg: ① $\delta(p, a, A) \ni (q, \Lambda)$.

// Pop stack top A
// out while reading
// input a and moving
// from state p to state q.



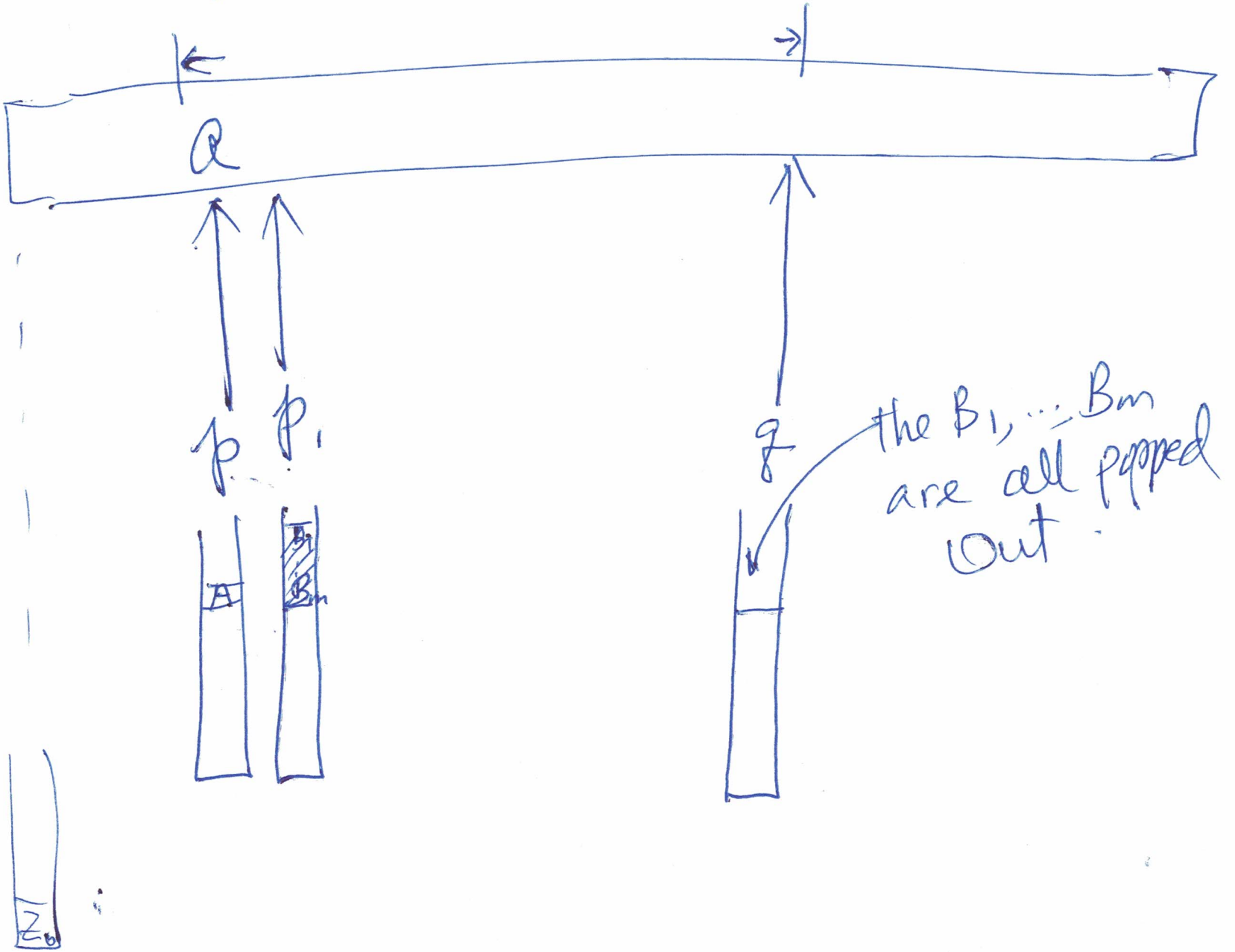
$[p, A, q] \longrightarrow a$

② . $\delta(p, a, A) \ni (p_1, \gamma)$ with $\gamma \neq \Lambda$;



let $\gamma = B_1 \ldots B_m$.

Recall: $\gamma = B_1 \cdots B_m$,



$$[p, A, q] \longrightarrow a[p_1, B_1, p_2][p_2, B_2, p_3] \cdots [p_{m-1}, B_m, p_m]$$

where, $p_m = q$, where $\gamma = B_1 \cdots B_m$.

③ Initial rule in G:

$$S \longrightarrow [q_0, Z_0, q].$$

Next time: Example.